



The Effect of Adding Clonidine to Articaine and Epinephrine on Post-treatment Pain: A Randomized Clinical Trial Study

Elham Shadmehr^{a*} , Sara Hashemi^b , Seyed Saman Hashemi^c , Yoo Jung Chung^d ,
Arman Goudarzi^c , Abbasali Khademi^e

^a Department of Endodontics and Periodontics, Division of Endodontics, School of Dentistry, University of California, San Francisco, USA; ^b Dental Students' Research Committee, School of Dentistry, Isfahan University of Medical Sciences, Isfahan, Iran; ^c Dentists, Private Practice, Isfahan, Iran; ^d School of Dentistry, University of California, San Francisco, USA; ^e Dental Research Center, Department of Endodontics, Dental School, Isfahan University of Medical Sciences, Isfahan, Iran

ARTICLE INFO

Article Type: Clinical Trial

Received: 25 Apr 2021

Revised: 26 Jul 2021

Accepted: 19 Aug 2021

Doi: 10.22037/iej.v16i4.33555

*Corresponding author: Elham Shadmehr, UCSF Preventive & Restorative Dental Science, School of Dentistry, 707 Parnassus Avenue, D3214, San Francisco, CA 94143-0758, USA.

Tel: +41-54765802

E-mail: elham.shadmehr@ucsf.edu

Introduction

An essential part of root canal therapy is to prevent and manage post-operative pain [1], which may persist after treatment [2, 3]. However, acquiring adequate pain control has been challenging as the frequency of post-operative pain after endodontic treatment is reported to be 3-58% [4]. Ways to achieve profound analgesia after root canal treatment are therefore imperative for the patient [5]. Currently, local anesthetic drugs, such as lidocaine, articaine, prilocaine, mepivacaine and bupivacaine, are commonly used to reduce pain during dental procedures [6]. In 2000, 4% articaine with

epinephrine was approved for dental use by the Food and Drug Administration in the United States [7]. Articaine is an amide anesthetic containing a lipophilic part, connected to a hydrophilic part. Although the thiophene ring in the lipophilic part differentiates articaine from other amide anesthetics; many studies have suggested that this structure causes higher neurotoxicity [8]. Pharmacokinetics of articaine leads to a fast onset and a short-acting pulpal anesthesia in inferior alveolar nerve block (IANB), compared to lidocaine [9]. In recent years, more adjunctive drugs have been used to increase analgesia during and after dental procedures, and the combinatorial use of medications has been advocated [10].

Clonidine is an α_2 -adrenoceptor agonist. It is used as an adjunct to enhance the anesthetic efficacy and induce greater analgesia in various injections such as dentoalveolar [11], spinal [12], epidural [13], brachial plexus, and peripheral nerve block anesthesia [14]. A previous study suggested that addition of a low dose of clonidine to bupivacaine in the supraclavicular brachial plexus block significantly prolonged the analgesic duration without producing any significant clinical complications [15]. Three other studies also reported that the adjunctive use of clonidine to lidocaine reduced pain after inferior alveolar nerve block (IANB) [16, 17], and substantially increased the success rate of anesthesia [11]. There are reports that recommend the addition of clonidine to articaine/epinephrine local infiltration anesthesia in pediatric dental setting due to the great increase in depth and duration of anesthesia [18].

Although both articaine and clonidine have been extensively studied in the past, a review of the literature revealed that there has been no study to date on the effect of clonidine in combination with articaine/epinephrine on post-operative pain in endodontics. Therefore, wished to determine the effect of adding clonidine to articaine/epinephrine on post-operative pain relief in patients with irreversible pulpitis.

Materials and Methods

Patient recruitment

Our double-blind clinical trial was conducted in Isfahan Dental School with a total of 105 patients who provided informed consent. All included patients were aged 15-60 years and healthy (Grade I, according to the classification of American Society of Anesthesiologists). Patients with underlying diseases such as diabetes, hypertension or other cardiovascular diseases or contraindication to clonidine, articaine, or epinephrine were not included in the study. In order to be included in the study, they were required to have a symptomatic irreversible pulpitis. The clinical diagnosis was based on subjective and objective findings indicating that the vital inflamed pulp is incapable of healing and the tooth exhibiting intermittent or spontaneous pain in a mandibular posterior tooth [19].

Following isolation with cotton rolls, and drying the tooth with air syringe, an endodontist, who was not involved in the study, performed a cold test to confirm the diagnosis. He used Endo Ice (1,1,1,2 tetrafluoroethane; Hygenic Corp, Akron, OH, USA) on a cotton pellet and covered it for maximum of 15 sec on the middle third of the buccal surface of the tooth until the patient responded. Any exaggerated reaction to cold stimulant & lingering of discomfort for more than 10 sec or more after removal of stimulant

was recorded. Those with no response, periodontal problems or active signs of oral infection or inflammation were excluded. Their periosteal bone and periradicular tissues were evaluated radiographically. None of the selected patients showed any pathology on their radiographs, other than widened periodontal ligament space. Patients with clinical and radiographic signs of pulpal necrosis or non-vital coronal pulp tissue upon access cavity preparation (partial necrosis) were excluded.

Patients who required multiple dental treatments or had consumed opioids during the week before or 72 h after the procedure were excluded from the study. Pregnant or lactating mothers, drug abusers, and β -blocker users were also excluded from the study. Patients who did not cooperate in pain assessment or needed additional anesthetic injections due to block failure were also excluded. Those with a changed treatment plan were excluded as well. After approval by the Institutional Ethics Committee of Isfahan Dental School (ID: IR.MUI.REC.1396.3.272), the study was registered in the Iranian Registry of Clinical Trials (ID: IRCT20201031049199N1). After an oral explanation of the procedure and possible complications was given, written informed consent was obtained from all patients. To ensure transparency and quality, the CONSORT (Consolidated Standards of Reporting Trials) checklist was followed (Figure 1).

Preparation of anesthetic solutions

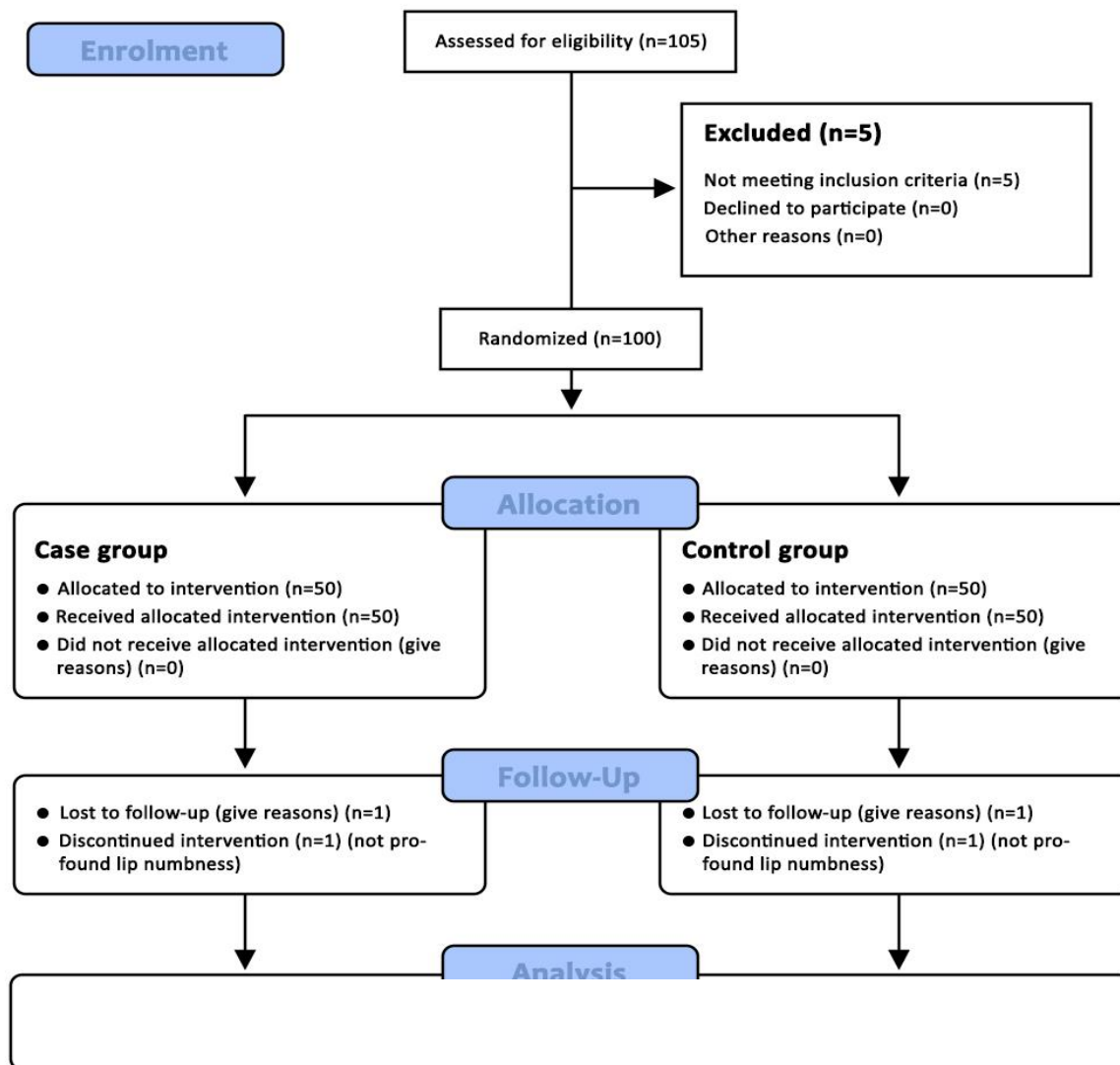
In order to make the study double-blind, the anesthetic solutions were prepared and encoded by a dental student who was not involved in the study. Using a 30-gauge insulin syringe (Keltenstrabe, Tuttlingen, Germany), either clonidine or distilled water was inserted into a cartridge of 4% articaine with 1:100,000 epinephrine, which increased the total volume of each vial content to 2 mL. The cartridges *with* and *without* clonidine were labelled No. 1 and No. 2, respectively. The solutions were prepared daily under aseptic conditions.

Randomization

An endodontist, who was blind to the contents of the cartridges, threw a dice just before treatment. No. 1 cartridge was used when an odd number came out and a No. 2 cartridge was used with an even number. Therefore, both the patient and the investigator were unaware of the type of local anaesthesia used. This randomization protocol can lead to a selection bias, but it was unavoidable due to limited time and financial sources.

Anesthesia

After diagnosis of irreversible pulpitis for the target tooth, a questionnaire including demographic information, medical history and pain intensity using a 170 mm Heft-Parker visual analog scale (VAS) was given to each patient [20]. The clonidine



group, cartridge No 1 ($n=50$), received an IANB with 0.2 mL clonidine (Catapressan amp, 150 $\mu\text{g/mL}$; Boehringer Ingelheim, Ingelheim am Rhein, Germany) in addition to 1.8 mL of 4% articaine with 1:100,000 epinephrine (Daropaksh, Tehran, Iran). The control group (No 2, $n=50$) received an IANB with 0.2 mL distilled water in addition to 1.8 mL 4% articaine with 1:100,000 epinephrine (Daropaksh, Tehran, Iran).

Topical anesthetic gel of 20% benzocaine (Denticare, Medicom, Canada) was applied at the IANB site for one min using a cotton swab. A standard IANB was performed with a 27-gauge 1.5-inch needle by one endodontist. After touching the medial border of the mandibular ramus (which located lateral to the pterygomandibular fold and the sphenomandibular ligament [21] from the top of the opposite side premolars, the

needle was withdrawn by 1 mm. After aspiration, the anesthetic solution was deposited for 2 min. If profound lip numbness was not obtained within 15 min, the block was considered failed and the patient was excluded from the study. For each patient, blood pressure and heart rate were measured before and 5 min [11] after the anesthetic administration using a digital blood pressure monitor (Omron M3; Omron Healthcare Co., Kyoto, Japan). The treatment was done in a single-visit using RaCe rotary system (FKG Dentaire, La-Chaux-de Fonds, Switzerland) according to the manufacturer instructions [22, 23]. Finally, the access cavity was sealed with eugenol temporary dressing (Zonalin; Associated Dental Products, Wiltshire, United Kingdom). The occlusion was checked to remove any occlusal interference.

Evaluation of post-operative pain

We evaluated the post-operative pain with HP VAS which was given to each patient. VAS is divided into 4 categories (no pain: 0 mm; mild pain: > 0 mm and ≤ 54 mm; moderate pain: > 54 mm and < 114 mm; severe pain: ≥ 114 mm). Patients were followed up via phone calls or text messages at 6, 12, 24, 48 and 72 h after the root canal treatment. All participants were asked about nausea, vomiting, drowsiness, dizziness and headaches. They were also asked about post-operative analgesic consumption and instructed to use up to 400 mg of ibuprofen (Gelofen; Jaber Ebne Hayyan Pharmaceutical Mfg. Co., Tehran, Iran) in case of an emergency pain [24]. The few that did have to consume analgesia post-operatively (only one person in clonidine group), were excluded in data analysis in order to prevent calculation error.



Acknowledgment

This study is a thesis project for general dentistry, funded by the School of Dentistry, Isfahan University of Medical Sciences, Isfahan, Iran.

Conflict of Interest: 'None declared'.

References

1. Gotler M, Bar-Gil B, Ashkenazi M. Postoperative pain after root canal treatment: a prospective cohort study. *Int J Dent*. 2012;2012:310467.
2. Law AS, Durand EU, Rindal DB, Nixdorf DR. "Doctor, why does my tooth still hurt"? *Northwest Dent*. 2010;89(1):33-6.
3. Sadaf D, Ahmad MZ. Factors associated with postoperative pain in endodontic therapy. *Int J Biomed Sci*. 2014;10(4):243-7.
4. Sathorn C, Parashos P, Messer H. The prevalence of postoperative pain and flare-up in single- and multiple-visit endodontic treatment: a systematic review. *Int Endod J*. 2008;41(2):91-9.
5. Chou R, Gordon DB, de Leon-Casasola OA, Rosenberg JM, Bickler S, Brennan T, Carter T, Cassidy CL, Chittenden EH, Degenhardt E, Griffith S, Manworren R, McCarberg B, Montgomery R, Murphy J. Management of Postoperative Pain: A Clinical Practice Guideline From the American Pain Society, the American Society of Regional Anesthesia and Pain Medicine, and the American Society of Anesthesiologists' Committee on Regional Anesthesia, Executive Committee, and Administrative Council. *J Pain*. 2016;17(2):131-57.
6. Allegretti CE, Sampaio RM, Horliana AC, Armonia PL, Rocha RG, Tortamano IP. Anesthetic Efficacy in Irreversible Pulpitis: A Randomized Clinical Trial. *Braz Dent J*. 2016;27(4):381-6.
7. Paxton K, Thome DE. Efficacy of articaine formulations: quantitative reviews. *Dent Clin North Am*. 2010;54(4):643-53.
8. Hopman AJG, Baart JA, Brand HS. Articaine and neurotoxicity – a review. *British Dental Journal*. 2017;223(7):501-6.
9. Mikesell P, Nusstein J, Reader A, Beck M, Weaver J. A comparison of articaine and lidocaine for inferior alveolar nerve blocks. *J Endod*. 2005;31(4):265-70.
10. Cecchetti MM, Negrato GV, Peres MP, Deboni MC, Naclério-Homem Mda G. Analgesic and adjuvant anesthetic effect of submucosal tramadol after mandibular third molar surgery. *Oral Surg Oral Med Oral Pathol Oral Radiol*. 2014;117(3):e249-54.
11. Shadmehr E, Aminozarbani MG, Akhavan A, Mahdavian P, Davoudi A. Anaesthetic efficacy of lidocaine/clonidine for inferior alveolar nerve block in patients with irreversible pulpitis. *Int Endod J*. 2017;50(6):531-9.
12. Jamali S, Monin S, Begon C, Dubousset AM, Ecoffey C. Clonidine in pediatric caudal anesthesia. *Anesth Analg*. 1994;78(4):663-6.
13. Singh M, Pathak A, Khan AL, Srivastava A, Negi D. Comparative Study of Clonidine with Ropivacaine versus Ropivacaine Alone in Epidural Anesthesia for Lower Limb Orthopedic Surgery. *Anesth Essays Res*. 2017;11(4):1035-9.
14. Hrishi AP, Rao G, Lionel KR. Efficacy of Clonidine as an Additive on the Duration of Action of Brachial Plexus Block Performed Under Ultrasound and Nerve Locator Guidance: A Prospective Randomized Study. *Anesth Essays Res*. 2019;13(1):105-10.

Conclusion

Our randomized clinical trial study demonstrated effective reduction in post-operative pain in patients who presented in pain and underwent endodontic treatment when compared with control. The addition of 30 µg clonidine to 1.8 mL 4% articaine with 1:100,000 epinephrine solution in IANB anesthesia during root canal and other dental treatments should be seriously considered and further studied. Our low dose of clonidine did not cause any adverse complications including hypotension, drowsiness, dizziness, nausea, vomiting and headache.

15. Chakraborty S, Chakrabarti J, Mandal MC, Hazra A, Das S. Effect of clonidine as adjuvant in bupivacaine-induced supraclavicular brachial plexus block: A randomized controlled trial. *Indian J Pharmacol.* 2010;42(2):74-7.
16. Brkovic B, Todorovic L, Stojic D. Comparison of clonidine and epinephrine in lidocaine anaesthesia for lower third molar surgery. *Int J Oral Maxillofac Surg.* 2005;34(4):401-6.
17. Patil PM, Patil SP. Is clonidine an adequate alternative to epinephrine as a vasoconstrictor in patients with hypertension? *J Oral Maxillofac Surg.* 2012;70(2):257-62.
18. Mel'nikova AV, Shugailov IA, Garus Ia N. [Efficiency of teeth local anesthesia by articaine-containing formulation with adrenaline and clonidine in pediatric dentistry]. *Stomatologiya (Mosk).* 2014;93(2):43-6.
19. Stefanac SJ. 8 - Acute phase of treatment. In: Stefanac SJ, Nesbit SP, editors. *Diagnosis and Treatment Planning in Dentistry* (Third Edition). St. Louis (MO): Mosby; 2017. pp. 173-91.e2.
20. Sirintawat N, Sawang K, Chaiyasamut T, Wongsirichat N. Pain measurement in oral and maxillofacial surgery. *J Dent Anesth Pain Med.* 2017;17(4):253-63.
21. Herring Fa. *Illustrated Anatomy of the Head and Neck.* 2012:216.
22. Hosseini HR, Parirokh M, Nakhaee N, P VA, Samani S. Efficacy of Articaine and Lidocaine for Buccal Infiltration of First Maxillary Molars with Symptomatic Irreversible Pulpitis: A Randomized Double-blinded Clinical Trial. *Iran Endod J.* 2016;11(2):79-84.
23. Sakhaeimanesh V, Khazaei S, Kaviani N, Saatchi M, Shafiei M, Khademi A. Anesthetic Efficacy of Articaine and Ketamine for Inferior Alveolar Nerve Block in Symptomatic Irreversible Pulpitis: A Prospective Randomized Double-Blind Study. *Iran Endod J.* 2017;12(4):449-53.
24. Talebzadeh B, Nezafati S, Rahimi S, Shahi S, Lotfi M, Ghasemi N. Comparison of manual and rotary instrumentation on postoperative pain in teeth with asymptomatic irreversible pulpitis: A randomized clinical trial. *Iran Endod J.* 2016;11(4):273.
25. Carroll I, Mackey S, Gaeta R. The role of adrenergic receptors and pain: The good, the bad, and the unknown. *Seminars in Anesthesia, Perioperative Medicine and Pain.* 2007;26(1):17-21.
26. Kuraishi Y, Hirota N, Sato Y, Kaneko S, Satoh M, Takagi H. Noradrenergic inhibition of the release of substance P from the primary afferents in the rabbit spinal dorsal horn. *Brain Res.* 1985;359(1-2):177-82.
27. Neil MJ. Clonidine: clinical pharmacology and therapeutic use in pain management. *Curr Clin Pharmacol.* 2011;6(4):280-7.
28. Yavari HR, Jafari F, Jamloo H, Hallaj-Nezhadi S, Jafari S. The Effect of Submucosal Injection of Corticosteroids on Pain Perception and Quality of Life after Root Canal Treatment of Teeth with Irreversible Pulpitis: A Randomized Clinical Trial. *J Endod.* 2019;45(5):477-82.
29. Kreimer T, Kiser R, 2nd, Reader A, Nusstein J, Drum M, Beck M. Anesthetic efficacy of combinations of 0.5 mol/L mannitol and lidocaine with epinephrine for inferior alveolar nerve blocks in patients with symptomatic irreversible pulpitis. *J Endod.* 2012;38(5):598-603.
30. Bigby J, Reader A, Nusstein J, Beck M. Anesthetic efficacy of lidocaine/meperidine for inferior alveolar nerve blocks in patients with irreversible pulpitis. *J Endod.* 2007;33(1):7-10.
31. Chowdhury S, Singh M, Shah A. Efficacy of lignocaine with clonidine and adrenaline in minor oral surgical procedure. *Contemp Clin Dent.* 2012;3(2):227-9.
32. Oertel R, Rahn R, Kirch W. Clinical pharmacokinetics of articaine. *Clin Pharmacokinet.* 1997;33(6):417-25.
33. Malet A, Faure MO, Deletage N, Pereira B, Haas J, Lambert G. The comparative cytotoxic effects of different local anesthetics on a human neuroblastoma cell line. *Anesth Analg.* 2015;120(3):589-96.
34. Baroni DB, Franz-Montan M, Cogo K, Berto LA, Volpato MC, Novaes PD, Groppo FC. Effect of articaine on mental nerve anterior portion: histological analysis in rats. *Acta Odontol Scand.* 2013;71(1):82-7.
35. Katyal V. The efficacy and safety of articaine versus lignocaine in dental treatments: a meta-analysis. *J Dent.* 2010;38(4):307-17.
36. Brandt RG, Anderson PF, McDonald NJ, Sohn W, Peters MC. The pulpal anesthetic efficacy of articaine versus lidocaine in dentistry: a meta-analysis. *J Am Dent Assoc.* 2011;142(5):493-504.
37. Kung J, McDonagh M, Sedgley CM. Does Articaine Provide an Advantage over Lidocaine in Patients with Symptomatic Irreversible Pulpitis? A Systematic Review and Meta-analysis. *J Endod.* 2015;41(11):1784-94.

Please cite this paper as: Shadmehr E, Hashemi S, Hashemi SS, Chung YJ, Arman Goudarzi, Khademi A. The Effect of Adding Clonidine to Articaine and Epinephrine on Post-treatment Pain: A Randomized Clinical Trial Study. *Iran Endod J.* 2021;16(4): 210-6. *Doi:* 10.22037/iej.v16i4.33555.